

Marine Petrosyan  
19.04..2026

# Coffee Shop Analytics: Spatial & Transactional Insights

---

PostgreSQL + PostGIS

# Project Inspiration: From Raw Data to Insight

**Challenge:** The `coffee_shop_raw` table was *fully denormalized*, mixing transactions, stores, products, and metadata — making analysis repetitive and inefficient.

**Goal:** Transform this raw, unstructured dataset into a **normalized, relational model** that supports spatial and transactional analytics.

**Vision:** Build a reproducible, automated workflow revealing how NYC coffee shops perform across time, space, and product categories.

**Motivation:** To understand not just *what sells*, but *where*, *when*, and *why* — turning operational data into strategic insight.

# Analytical Questions Driving the Exploration

- **How many transactions** define the overall coffee shop activity?
- What is the **geographical distribution** of stores across NYC?
- What are the **sales trends** by store over time?
- Which **store location** contributes most to total revenue?
- Which **store serves more people per square kilometer**?
- What is the **sales trend** for the **top-selling category**?
- What are the **peak hours** of customer activity?

**1. Schema Design & Normalization**

- Created normalized tables for stores, categories, products, variants, and transactions.
- Added spatial hierarchy with PostGIS polygons for NYC districts.

**2. Data Dictionary & ER Diagram**

- Documented every table and column with constraints.
- Generated ER diagram to visualize relationships.

**3. ETL Automation**

- Built Python + psycopg2/SQLAlchemy scripts.
- Automated refresh of analytical tables via Docker.

**4. Analytical Tables & Indexing**

- Created reproducible summary tables (sales by store, category, daily trends, etc.).
- Designed composite indexes for query optimization.

**5. Analytical Queries**

- Sales by store/day, category trends, product variant performance.
- Spatial queries (ST\_Area, ST\_Within, ST\_Intersects).
- Peak hours, basket analysis, store rankings.

**6. Performance Optimization**

- Benchmarked queries with EXPLAIN ANALYZE.
- Demonstrated improvements with composite and spatial indexes.

**7. Visualization & Insights**

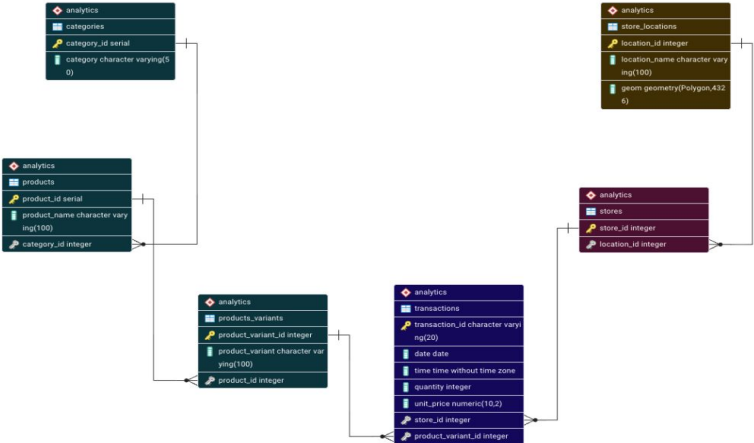
- Exported GeoJSON for heatmaps.
- Prepared dashboards in Google Slides with charts and maps.

# Data Dictionary Summary

- **store\_locations**: spatial polygons for NYC districts
- **stores**: store IDs linked to locations
- **products, categories, variants**: product hierarchy
- **transactions**: sales data with timestamps
- Analytical tables

# The schema integrates spatial and transactional data for NYC coffee shops, enabling advanced analytics on store performance, product trends, and geographic sales density.

|                                       |
|---------------------------------------|
| analytics                             |
| coffee_shop_raw                       |
| transaction_id character varying(20)  |
| date date                             |
| time time without time zone           |
| quantity integer                      |
| store_id integer                      |
| store_location character varying(100) |
| product_id integer                    |
| unit_price numeric(10,2)              |
| category character varying(50)        |
| product_name character varying(100)   |
| product_detail character varying(100) |



|                         |
|-------------------------|
| analytics               |
| daily_sales             |
| date date               |
| total_sales numeric     |
| total_units_sold bigint |
| num_transactions bigint |

|                     |
|---------------------|
| analytics           |
| hourly_sales        |
| hour numeric        |
| total_sales numeric |

|                            |
|----------------------------|
| analytics                  |
| price_trends               |
| product_variant_id integer |
| date date                  |
| avg_price numeric          |

|                                |
|--------------------------------|
| analytics                      |
| sales_by_category              |
| category_id integer            |
| category character varying(50) |
| total_sales numeric            |
| total_units_sold bigint        |

|                                      |
|--------------------------------------|
| analytics                            |
| sales_by_store                       |
| store_id integer                     |
| location_name character varying(100) |
| total_sales numeric                  |
| num_transactions bigint              |
| total_units_sold bigint              |

|                                      |
|--------------------------------------|
| analytics                            |
| sales_density                        |
| location_id integer                  |
| location_name character varying(100) |
| sales_per_m2 double precision        |

|                                      |
|--------------------------------------|
| analytics                            |
| store_location_summary               |
| location_id integer                  |
| location_name character varying(100) |
| geom geometry(Polygon,4326)          |
| total_sales numeric                  |
| total_units_sold bigint              |

|                         |
|-------------------------|
| analytics               |
| store_rankings          |
| store_id integer        |
| total_units_sold bigint |
| total_revenue numeric   |
| quantity_rank bigint    |
| revenue_rank bigint     |

|                                     |
|-------------------------------------|
| analytics                           |
| top_products                        |
| product_id integer                  |
| product_name character varying(100) |
| total_sales numeric                 |
| total_units_sold bigint             |

# ETL Workflow

Load raw CSVs into staging tables

Convert WKT → geometry

Populate normalized tables

Automate with Docker Compose



# Data Overview

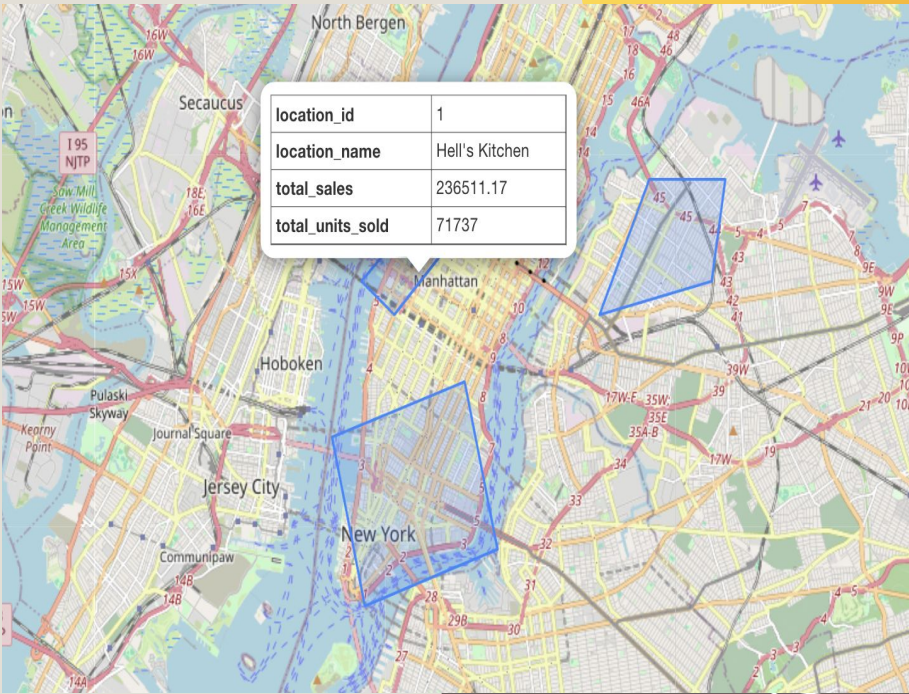
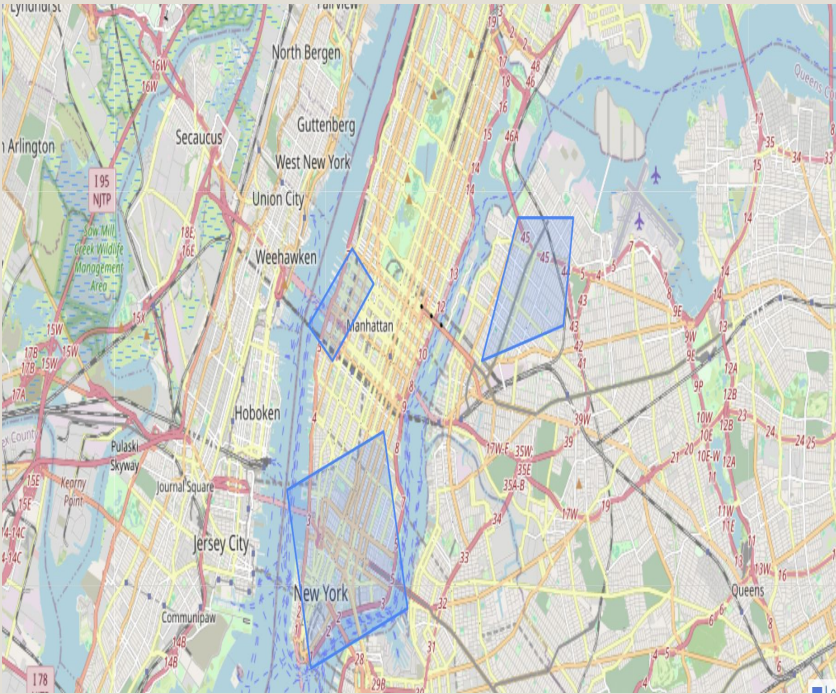
Over 149K transactions confirm robust customer activity across 3 stores and 9 categories.

## Transaction and Distinct Property Summary

| Transaction Summary |         |
|---------------------|---------|
| Total Transactions  | 149,116 |
| Unique Transactions | 149,116 |

| Distinct Property Overview |    |
|----------------------------|----|
| Stores                     | 3  |
| Store Locations            | 3  |
| Categories                 | 9  |
| Product IDs                | 80 |
| Product Names              | 29 |
| Product Details            | 80 |

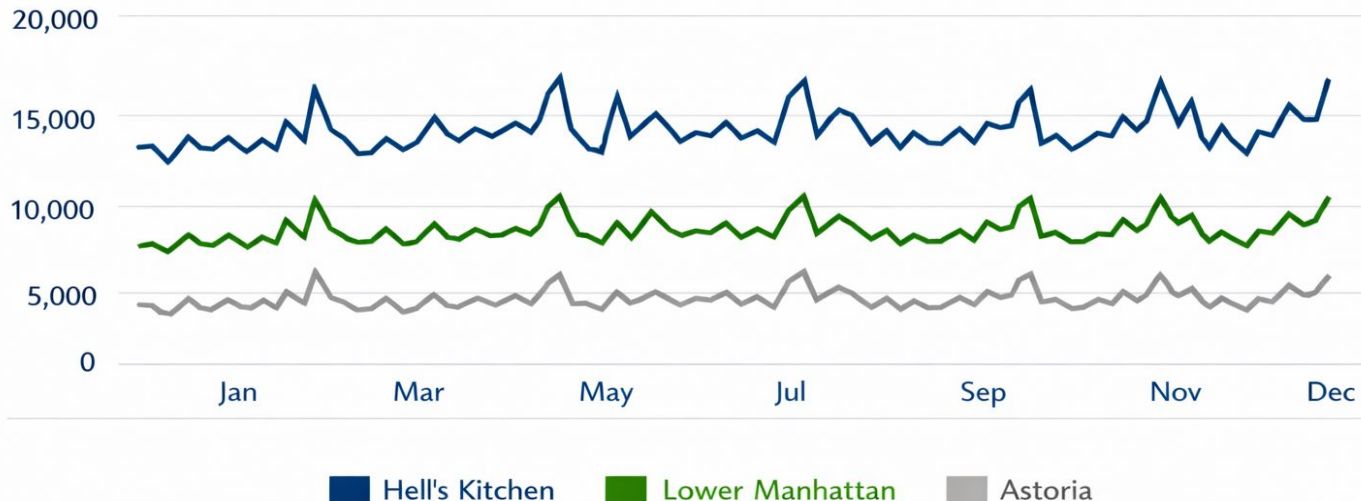
# Store Location Districts on the Map



## How varies revenue by district?

**All three shops show similar upward trends, with Hell's Kitchen leading revenue, Lower Manhattan fluctuating mid-range, and Astoria rising steadily.**

Store Sales Trends Over Time



# What are ranks of all stores by overall performance?

## Store Ranking

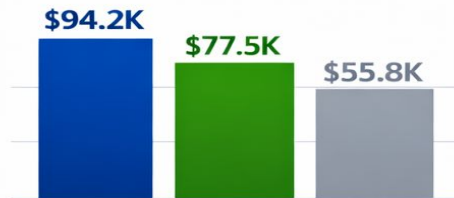
*Lower Manhattan leads, followed by Hell's Kitchen and Astoria.*

Quantity Sold Ranking



*Hell's Kitchen tops revenue, then Astoria and Lower Manhattan.*

Revenue Ranking



*Hell's Kitchen performs best overall.*

Overall Performance

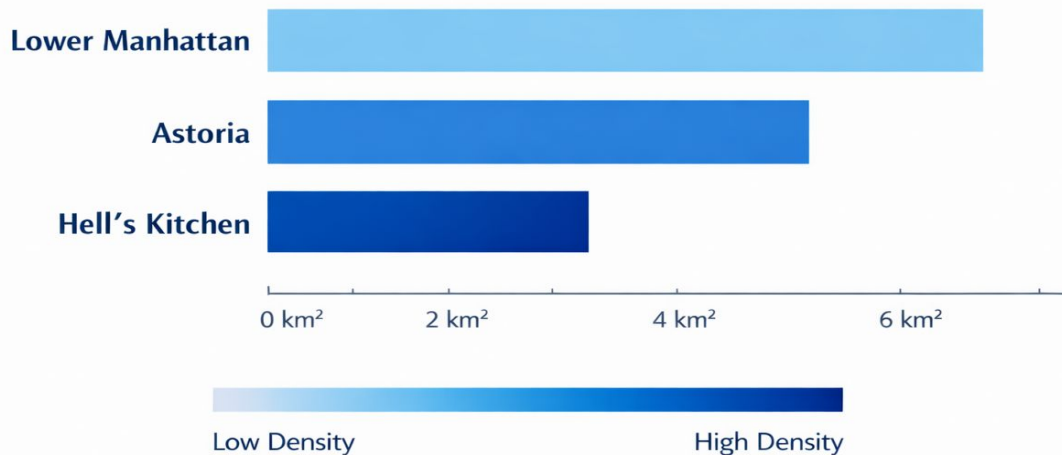
| Store           | Rank | Total Sales |
|-----------------|------|-------------|
| Hell's Kitchen  | 1st  | \$94,200    |
| Astoria         | 2ed  | \$77,500    |
| Lower Manhattan | 3rd  | \$55,800    |

Lower Manhattan Hell's Kitchen Astoria

# What is sales density by area?

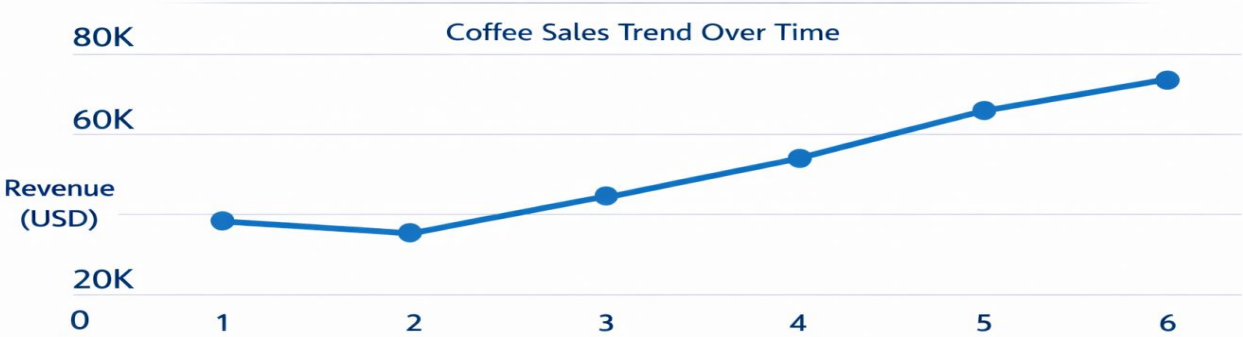
**Hell's Kitchen achieves the highest sales density,  
serving more people per square kilometer.**

Sales Density by Area



# How behaves Coffee revenue over time?

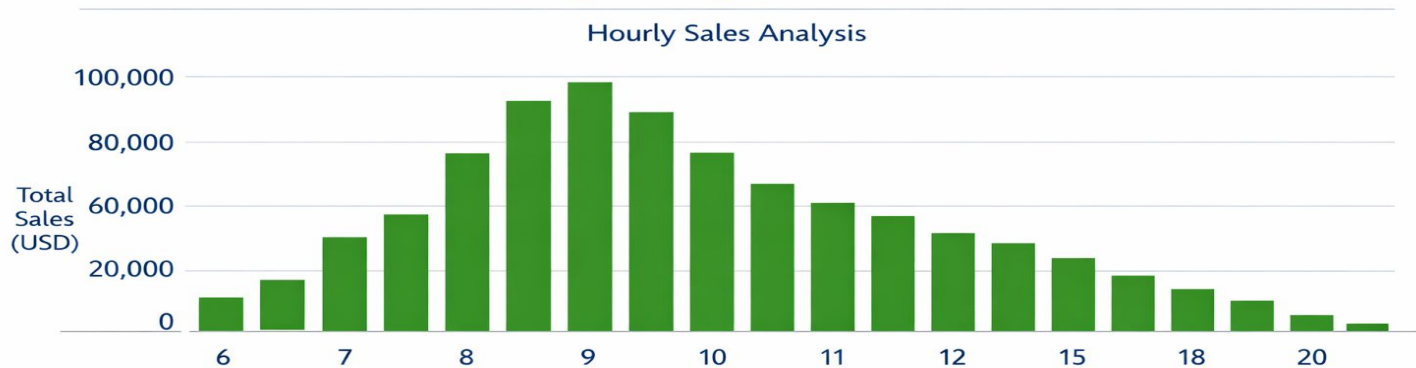
Coffee revenue rises steadily month to month, showing sustained growth through June.



| Month | Revenue (USD) |
|-------|---------------|
| 1     | \$31,256      |
| 2     | \$29,269      |
| 3     | \$38,304      |
| 4     | \$45,971      |
| 5     | \$60,363      |
| 6     | \$64,789      |

# What is customer flow throughout the day?

**Customer flow peaks between 8 AM and 10 AM, then declines steadily after noon—revealing morning operational burden.**



| Hour | Total Sales (USD) |
|------|-------------------|
| 10   | 88,673            |
| 9    | 85,170            |
| 8    | 82,700            |
| 7    | 63,526            |
| 11   | 46,319            |
| 15   | 41,733            |
| 16   | 41,123            |
| 19   | 22,936            |

# Key Insights from Coffee Shop Analytics

**Transaction Volume:** Over 149k transactions confirm robust customer activity.

**Store Comparison:** Hell's Kitchen leads in revenue, Lower Manhattan excels in quantity sold, Astoria shows balanced but developing performance.

**Product Trends:** Coffee sales show consistent month-to-month growth, confirming strong and sustained demand across the first half of the year.

**Temporal Patterns:** Revenue peaks at 8 AM and 10 AM, revealing morning operational burden.

**Spatial Context:** Lower Manhattan covers the largest area, but Hell's Kitchen achieves the highest sales density.

# Thank You!